4.7 Transformations of Polynomial Functions

Name _____ Period ____ Date _____

Algebra II 4.7 (30-minute plan) · Student sheet

I can…

- describe transformations of polynomial functions.
- write rules for transformed polynomial functions.
- graph transformed polynomial functions.

Vocabulary · fill in the missing word

Horizontal translation — _____

Vertical translation — _____

Reflections — $-f(x)$ in the _____; $f(-x)$ in the y-axis

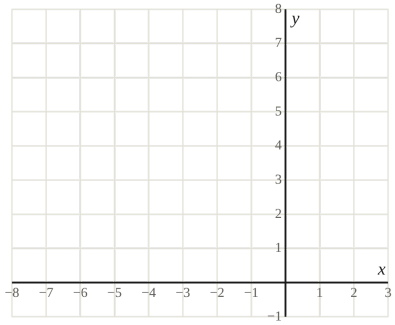
Stretches and shrinks — $a \cdot f(x)$ vertically; $f(ax)$ _____

Part 1 · Notes and examples (fill in each note, then work the example)

Translations: $f(x - h)$ moves the graph _____; $f(x) + k$ moves it _____.

Example 1

Describe the transformation of $f(x) = x^4$ represented by $g(x) = (x + 5)^4 + 1$.



Reflections: $f(-x)$ reflects in the _____; $-f(x)$ reflects in the _____.

Example 2

Let $f(x) = x^3 - 6$. Write a rule for g, a vertical stretch by a factor of 2 followed by a translation 3 units up.

Part 2 · Partner practice

Partner practice

Describe the transformation of $f(x) = x^3$ represented by $g(x) = (x - 4)^3 - 7$.